e) Caffeine

Caffeine taken during lactation may make a breastfed infant irritable and wakeful. During pregnancy caffeine consumption should be moderate say 1-2 cups of coffee a day. Larger doses of coffee may interfere with the availability of iron from the milk and impair the infant iron status.

2.3 Growth and development during infancy

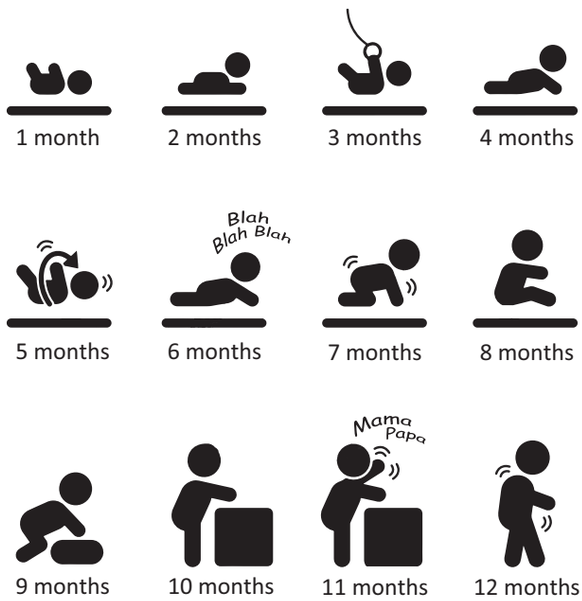


Fig 2.5 Stages of growth of an infant

Next to foetal period, the infants 1st year is the time of most rapid growth. According to WHO figures the average weight of most healthy new-born babies is around 3.2 kg. Healthy child doubles birth weight by six months. His weight becomes 3 times by the time the child is one year old. The normal birth length of 50 to 55 cm increases by another 23 to 25 cm during the first year. With increase in length the body proportions also change. At birth the child has 75% water 12 to 15 % fat. By the end of the one year the water content decreases to 60% and fat increases to 24 %. Muscles would be comparatively well developed at the end of the year. Infant have rapid heart rate, 120 - 140 per minute. But the haemoglobin level of well-nourished infant is 17 - 20 g per 100 ml. This provides a reserve for expansion of the blood circulation and adequate oxygen carrying capacity to the growing tissue during the first 4 - 6 months.

The full term infant is able to digest protein, emulsified fats and simple carbohydrates such as lactose. Salivary secretion and gastric acidity are low in infants under 3 months. Kidney reach their full functional capacity by the end of 1st year. During the first few months the glomerular filtration rate is somewhat lower and therefore the excretion of high concentration of solutes is more difficult. The increase in the number of brain cells is most rapid during foetal life and in the first five to six months after birth.

2.3.1 Nutritional requirements

a) Energy

Study reveal that basal and total energy requirement for infants are higher than adults per unit body weight. Infants require 92 Kcal / kg body weight. For one month old infant, 50 % energy intake is used for basal energy, 25 % for activity and 25 % for growth. Extremely active children may require up to 40% energy for activity. 70% of calories can be met by milk alone and rest of the calories have to be supplied by supplementary foods after six months.

b) Protein

Protein intake of a healthy infants is about 1.16 g/ kg body weight. The recommendations for infants are based on the total protein content and amino acid pattern of the average daily intake of human milk. Human milk provides all the amino acids more than the required amount needed for proper growth. Histidine, which is a non-essential amino acid for adults, is necessary for growth and maintenance of an infant. Human milk protein is 100% utilised. If calories and protein requirements are not met, infant suffer from protein energy malnutrition.

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